GPS Location Pinning

Lost Pet Alert System — Location & Distance Matching

PawGuard Platform | pawguard-backend (FastAPI + PostgreSQL + Redis + ARQ) | Prepared for Team Review

1. Purpose

Both lost and found reports capture a precise geographic pin (not just a free-text address), so the matching engine can score candidate pairs by real-world proximity instead of relying on address text alone.

2. Data Captured Per Report

Column	Type	Notes
location_address	TEXT	Human-readable address, always required
latitude	NUMERIC(9,6)	Optional — geocoded from address or device GPS
longitude	NUMERIC(9,6)	Optional — geocoded from address or device GPS

Present on both lost_reports and found_reports, and also on vet_clinics for map-based clinic search.

3. Distance Calculation — Haversine Formula

_calculate_distance_km() computes true great-circle distance between the two pins (Earth radius 6,371 km), not straight-line map distance, so results stay accurate over longer spans.

If either report is missing coordinates, distance defaults to 999 km (effectively “unknown / no distance credit”) rather than crashing the match.

4. Distance Contributes to the Match Score

Distance Band	Points Awarded (of 20 max)
≤ 2 km	20.0 — distance_near
≤ 5 km	12.0 — distance_close
≤ 10 km	6.0 — distance_moderate
> 10 km	0 — no distance credit

This is one of six weighted factors (see Lost Pet Alert System Overview) that together produce a 0–100 confidence_score stored on report_matches, alongside the raw distance_km value for transparency.

5. Also Used For

- Sorting/filtering lost & found reports near a user's current location.
- Vet clinic directory search radius (see Vet Directory doc) — same latitude/longitude pattern on vet_clinics.

Source: lost_found/service.py (_calculate_distance_km, _evaluate_match_score), lost_found/models.py.